

Wi-Fi Series

Data Sheet

USE_8800_LH

Wi-Fi / BLE Module

Version

VERSION	REVISION DATE	REVISOR	REVIEWER	REVISED CONTENTS
1.0	05/08/2026	MB	LB	Initial release

Contents

Features :	3
General.....	3
MCU Platform.....	3
WLAN Features.....	3
Host Interface.....	3
Bluetooth Features.....	3
General Description:	4
Pinout Diagram:	4
Pin Descriptions:	5
Electrical Specifications :	6
Power Supply DC Characteristics:	6
Power Consumption.....	6
RF Characteristics.....	7
1. 2.4G RF Specification.....	7
2. 5GHz RF Specification.....	8
3. Bluetooth Specification.....	9
Block Diagram :	10
Reference Design :	11
Recommended Footprint Dimensions:	12
Module Dimension Drawing:	14
Product Dimension:	14
Recommended Reflow Profile :	15

Features :

General

- Support Wi-Fi/Bluetooth functionalities
- Compliant with IEEE 802.11a/b/g/n/ac/ax
- Bluetooth 5.4 compliant
- Support Wi-Fi 6 TWT
- 1T1R architecture
- Support STBC, beamforming
- Dimension: 12 x 12 x 2.47 mm
- Operating temperature: -20°C to 80°C
- Storage temperature: -40°C to 125°C
- OS supported : Android /Linux

MCU Platform

- Chipset: AIC8800D40L-H

WLAN Features

- Support 2.4G+5G dual band, 20/40MHz bandwidth
- Support STA, SoftAP, Wi-Fi Direct modes concurrently
- Data rates up to 286.8Mbps with 20/40MHz bandwidth
- Wi-Fi Security WEP / WPA / WPA2/WPA3-SAE Personal, MFP

Host Interface

- Supports SDIO3.0 interface for WLAN
- Supports UART interface for Bluetooth
- Supports SDIO3.0/USB2.0

Bluetooth Features

- Supports all the mandatory and optional features of Bluetooth 5.4
- Supports advanced master and slave topologies
- Small SMT package

General Description:

The **USE_8800_LH** is a highly integrated chip with dual-band Wi-Fi 6 and Bluetooth 5.4 for wireless applications. Its WLAN function supports the SDIO 3.0 interface, and Bluetooth supports the UART interface only. The module provides a simple legacy and 20MHz/40MHz co-existence mechanism to ensure backward and network compatibility, and supports both 2.4GHz and 5GHz Wi-Fi 6 bands.

The wireless module complies with the IEEE 802.11a/b/g/n/ac/ax standard and supports data rates up to 286.8Mbps with 20/40MHz bandwidth operation.

For more information, contact us at info@unisemindia.com.

Pinout Diagram:

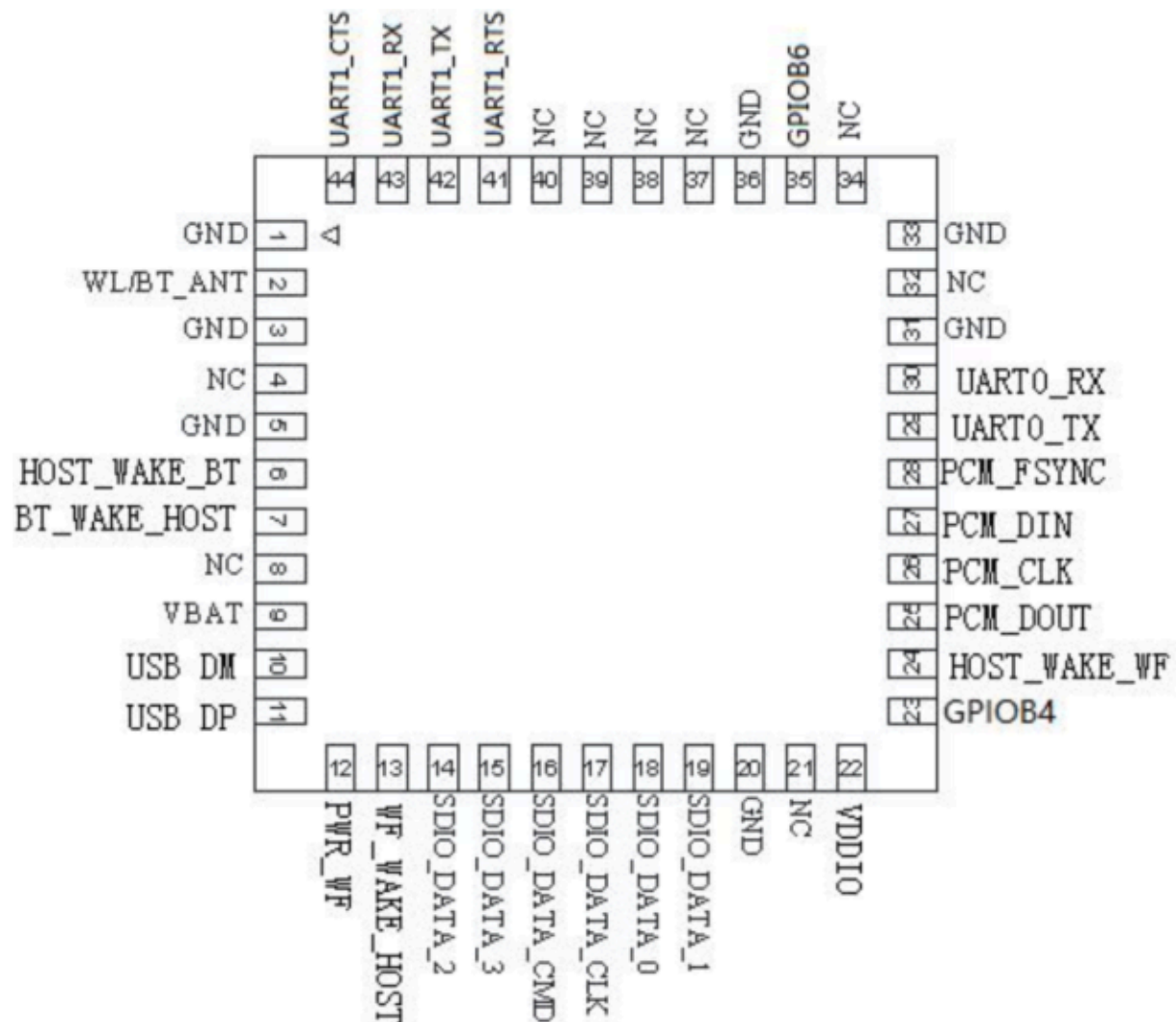


Fig1: Module Pinouts

Pin Descriptions:

NO.	Name	Type	Description	Voltage
1	GND	—	Ground connections	
2	WL/BT_ANT	I/O	WLAN and BT RF I/O port	
3	GND	—	Ground connections	
4	NC	—	Floating (Don't connected to ground)	
5	GND	—	Ground connections	
6	HOST_WAKE_BT	I/O	GPIOB3/HOST_WAKE_BT	VIO
7	BT_WAKE_HOST	I/O	GPIOB2/BT_WAKE_HOST	VIO
8	NC	—	Floating (Don't connected to ground)	
9	VBAT	P	Main power voltage source input	3.3V
13	WF Wake Up Host	I/O	WF Wake Up Host	VIO
14	SD_D2	I/O	SDIO data line 2, GPIOA15	VIO
15	SD_D3	I/O	SDIO data line 3, GPIOA14	VIO
16	SD_CMD	I/O	SDIO command line, GPIOA13	VIO
17	SD_CLK	I/O	SDIO clock line ,GPIOA12	VIO
18	SD_D0	I/O	SDIO data line 0 ,GPIOA11	VIO
19	SD_D1	I/O	SDIO data line 1, GPIOA10	VIO
20	GND	—	Ground connections	
21	NC	—	Floating (Don't connected to ground)	
22	VDDIO	P	I/O Voltage	1.8V/3.3V
23	GPIOB4	I/O	GPIOB4,if not used please keep NC	VIO
24	HOST_WAKE_WF	I/O	HOST_WAKE_WF	VIO
25	PCM_DOUT	O	PCM Data output	VIO
26	PCM_CLK	I/O	PCM clock	VIO
27	PCM_DIN	I	PCM data input	VIO
28	PCM_SYNC	I/O	PCM sync signal	VIO
29	UART0 TX	I/O	GPIOA9, UART0 TX	VIO
30	UART0 RX	I/O	GPIOA8, UART0 RX	VIO
31	GND	—	Ground connections	
32	NC	—	Floating (Don't connected to ground)	
33	GND	—	Ground connections	
34	NC	—	Floating (Don't connected to ground)	

35	GPIOB6	I/O	GPIOB6,,if not used please keep NC	VIO
36	GND	—	Ground connections	
37	NC	—	Floating (Don't connected to ground)	
38	NC	—	Floating (Don't connected to ground)	
39	NC	—	Floating (Don't connected to ground)	
40	NC	—	Floating (Don't connected to ground)	
41	UART1_RTS	I/O	BT_UART1_RTS, GPIOA7	VIO
42	UART1_TX	I/O	BT_UART1_TX, GPIOA5	VIO
43	UART1_RX	I/O	BT_UART1_RX, GPIOA4	VIO
44	UART1_CTS	I/O	BT_UART1_CTS, GPIOA6	VIO

The signal type codes are used in the table:

- P:POWER
- I:INPUT
- O:OUTPUT

Electrical Specifications :

Power Supply DC Characteristics:

Parameter	MIN	TYP	MAX	Unit
Operating Temperature	-20	25	80	°C
VBAT	3.0	3.3	3.6	V
VDDIO	1.7	1.8 / 3.3	3.6	V

Power Consumption

Test condition: VBAT = 3.3 V / VDDIO = 3.3 V

2.4 GHz

Operating Mode	Maximum TX Current (mA)	Maximum RX Current (mA)
11b @ 20 dBm	268	57
11g @ 17 dBm	187	57
HT20-MCS7 @ 16 dBm	179	58
HT40-MCS7 @ 16 dBm	163	57

VHT20-MCS8 @ 16 dBm	164	56
VHT40-MCS9 @ 16 dBm	162	56
HE20-MCS11 @ 15 dBm	138	57
HE40-MCS11 @ 15 dBm	134	58

5GHz

Operating Mode	Maximum TX Current (mA)	Maximum RX Current (mA)
54M @ 17 dBm	215	58
MCS7 HT20 @ 17 dBm	206	56
MCS7 HT40 @ 17 dBm	188	56
MCS8 VHT20 @ 17 dBm	203	58
MCS9 VHT40 @ 16 dBm	186	57
MCS11 HE20 @ 15 dBm	185	58
MCS11 HE40 @ 14 dBm	174	58

RF Characteristics

1. 2.4G RF Specification

Feature	Description	
WLAN Standard	IEEE 802.11b/g/n/ax, Wi-Fi compliant	
Frequency Range	2.400 GHz ~ 2.4835 GHz (2.4 GHz ISM Band)	
Number of Channels	2.4GHz : Ch1 ~ Ch13,Ch14	
Modulation	DBPSK/DQPSK/CCK(DSSS)、BPSK/QPSK/16QAM/64QAM(OF DM)	
Test Items	Typical Value	EVM
Output Power	802.11b /11Mbps : 18dBm $\pm$ 2 dBm	EVM -10dB
	802.11g /54Mbps : 15dBm $\pm$ 2 dBm	EVM -25dB
	802.11n /MCS7: 15dBm $\pm$ 2 dBm	EVM -28dB
	802.11ax /MCS9 :14dBm $\pm$ 2 dBm	EVM -32dB
	802.11ax /MCS11 :13dBm $\pm$ 2 dBm	EVM -35dB

Spectrum Mask	Meet with IEEE standard	
Freq. Tolerance	±20ppm	
ReceiveSensitivity (11b,20MHz) @8% PER	- 11Mbps@-85dBm	≤-76dBm
ReceiveSensitivity (11g,20MHz) @10% PER	- 54Mbps@-73dBm	≤-68dBm
Receive Sensitivity (11n,20MHz) @10% PER	- MCS=7 @-70dBm	≤-67dBm
Receive Sensitivity (11n,40MHz) @10% PER	- MCS=7 @-68dBm	≤-64dBm
Receive Sensitivity (11ax,20MHz) @10% PER	- MCS=11@-63dBm	≤-52dBm
Receive Sensitivity(11ax,40MHz) @10% PER	- MCS=11@61-dBm	≤-49dBm
Maximum Input Level	802.11b : -10 dBm	
	802.11g/n : -20 dBm	
Antenna Reference	Small antennas with 0~2 dBi peak gain	

2. 5GHz RF Specification

Feature	Description	
WLAN Standard	IEEE 802.11 a/n Wi-Fi compliant	
Frequency Range	5.150 GHz ~ 5.850 GHz (5.0 GHz Band)	
Test Items	Typical Value	EVM
Output Power	802.11a 54Mbps: 15dBm ± 2 dBm	EVM -25dB
	802.11n MCS7: 15 dBm± 2 dBm	EVM -28dB
	802.11ac MCS8: 14dBm ± 2 dBm	EVM -30dB
	802.11ac MCS9: 14dBm ± 2 dBm	EVM -32dB
	802.11ax MCS11 : 12dBm ± 2 dBm	EVM -35dB
ReceiveSensitivity (11a,20MHz) @10% PER	- 6Mbps PER @ -89 dBm, typical	≤-85
	- 54MbpsPER @ -71 dBm, typical	≤-68
Receive Sensitivity (11n,20MHz) @10% PER	- MCS=0 PER @ -89 dBm, typical	≤-85
	- MCS=7 PER @ -69 dBm, typical	≤-67

Receive Sensitivity (11n,40MHz) @10% PER	- MCS=0 PER @ -87 dBm, typical	≤-82
	- MCS=7 PER @ -67 dBm, typical	≤-64
Receive Sensitivity (11ac,20MHz) @10% PER	- MCS=0 PER @ -85 dBm, typical	≤-81
	- MCS=8 PER @ -66 dBm, typical	≤-59
Receive Sensitivity (11ac,40MHz) @10% PER	- MCS=0 PER @ -83 dBm, typical	≤-79
	- MCS=9 PER @ -63 dBm, typical	≤-54
Receive Sensitivity (11ax,20MHz) @10% PER	- MCS=0 PER @ -82 dBm, typical	≤-78
	- MCS=11PER @ -65 dBm, typical	≤-52
Receive Sensitivity (11ax,40MHz) @10% PER	- MCS=0 PER @ -81 dBm, typical	≤-78
	- MCS=11PER @ -62 dBm, typical	≤-49
Maximum input level	802.11a/n: -30 dBm	
	802.11ac: -30 dBm	
	802.11ax: -30 dBm	

Conditions : VBAT=3.3V ; VDDIO=3.3V ; Temp:25°C

3. Bluetooth Specification

Feature	Description
General Specification	
Bluetooth Standard	LE(1Mbps, LE(2Mbps)
Host Interface	UART
Frequency Band	2400 MHz ~ 2483.5 MHz
Number of Channels	40 channels for BLE
Modulation	GFSK
RF Specification	
Output Power , tolerance ± 3 dB	
	CL1(dBm)
BDR Output Power	/
EDR Output Power	/

BLE Output Power	9
Sensitivity, tolerance :/ dB	
Sensitivity @ BER=0.1% for GFSK (1Mbps)	/
Sensitivity @ BER=0.01% for $\pi/4$ -DQPSK (2Mbps)	/
Sensitivity @ BER=0.01% for 8DPSK (3Mbps)	/
Sensitivity @ BER=30.8% for LE (1Mbps)	-85
Sensitivity @ BER=30.8% for LE (2Mbps)	-83
Maximum Input Level	GFSK :-20dBm

Block Diagram :

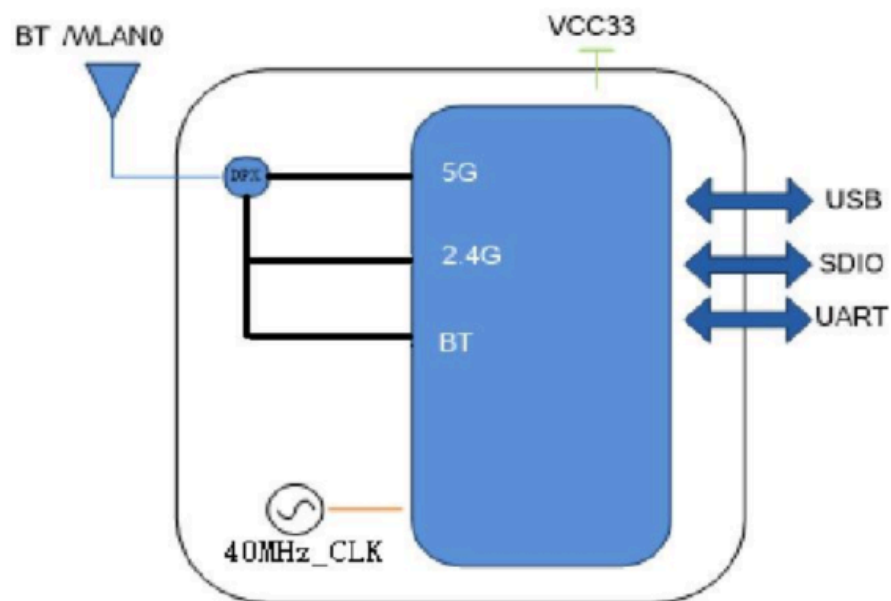


Fig 2 : Block Diagram

Reference Design :

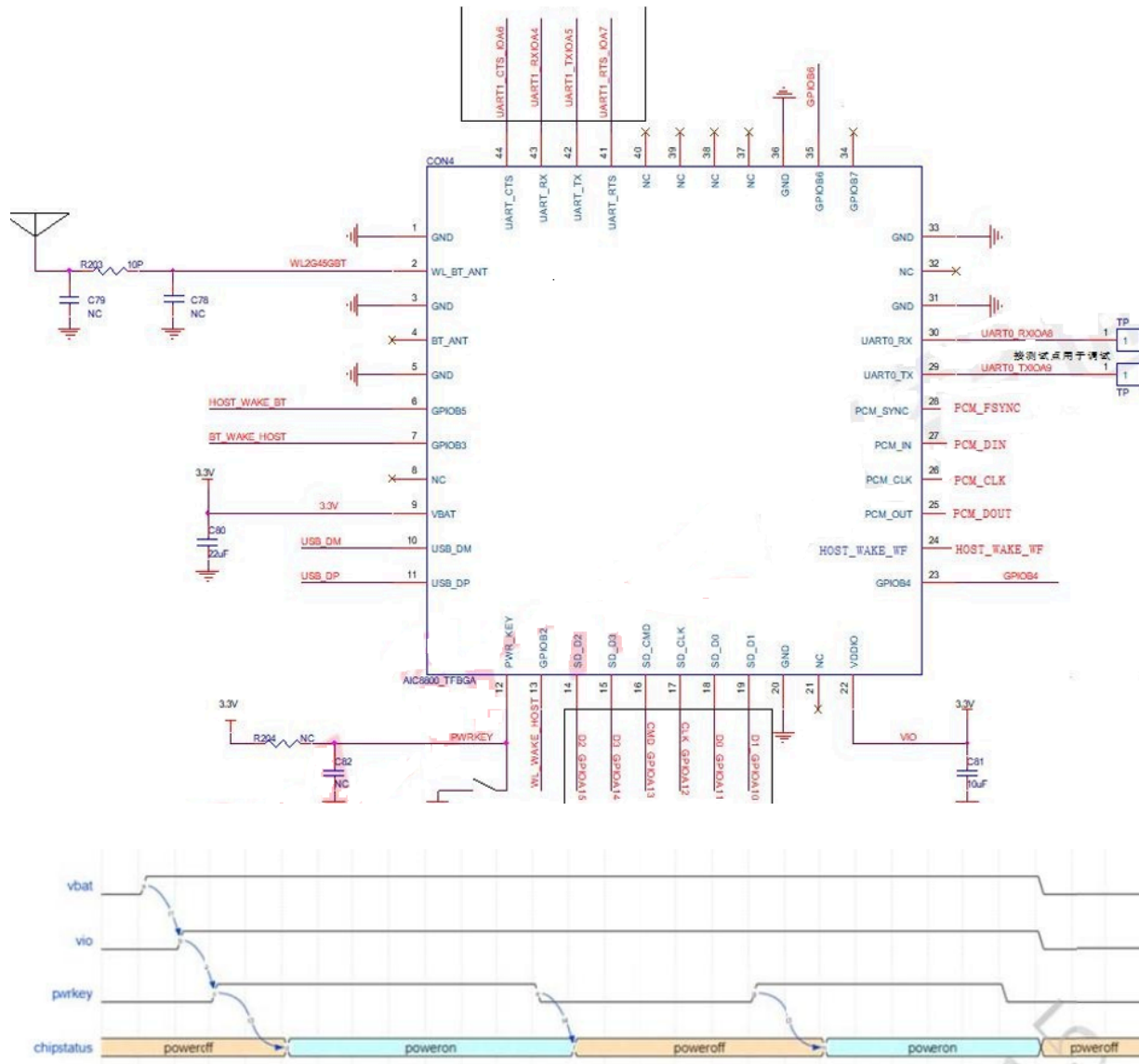


Fig 3 : Reference Design

- t1: VIO's power on time $\geq$ VBAT's
- t2: Power key's high time $\geq$ VIO's +200us
- t3: Chip all power on ready time $\geq$ power key high time + 8ms
- t4: Power key pull low to chip all power off time $\geq$ 6ms

Recommended Footprint Dimensions:

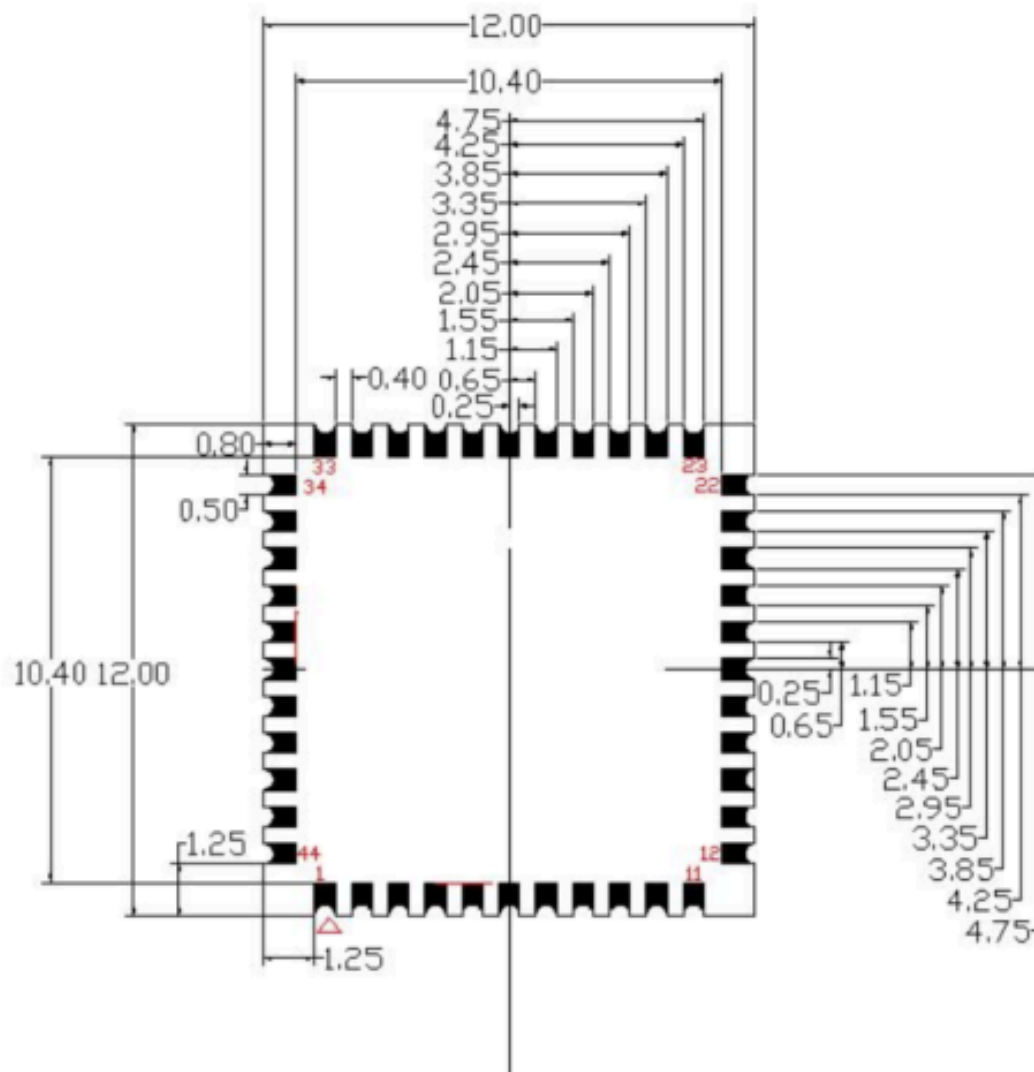


Fig 4(a) : Recommended Footprint Dimensions

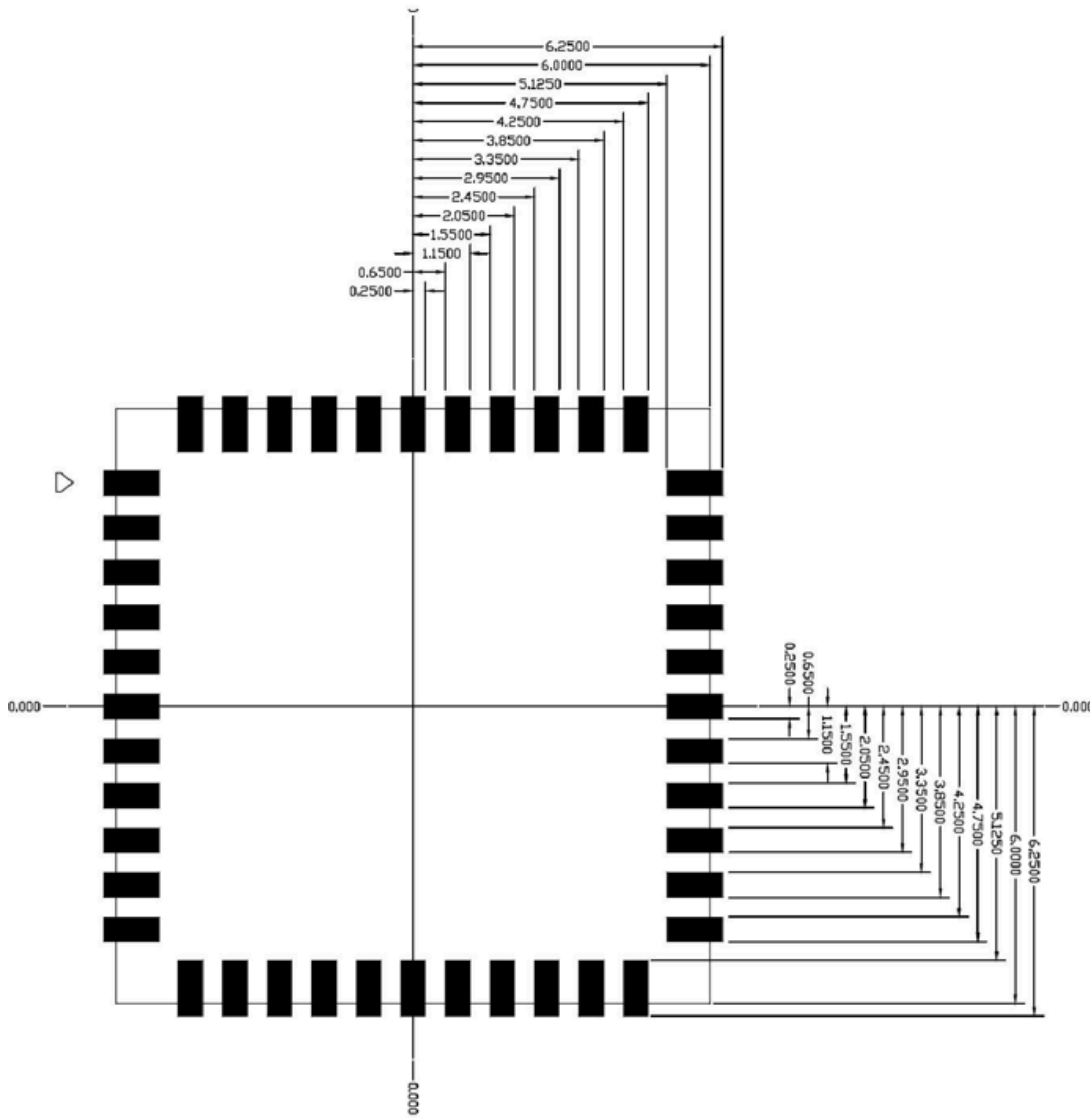


Fig 4(b) : Recommended Footprint Dimensions

Module Dimension Drawing:

- Module Dimension: L x W :12 x 12 (± 0.25) mm

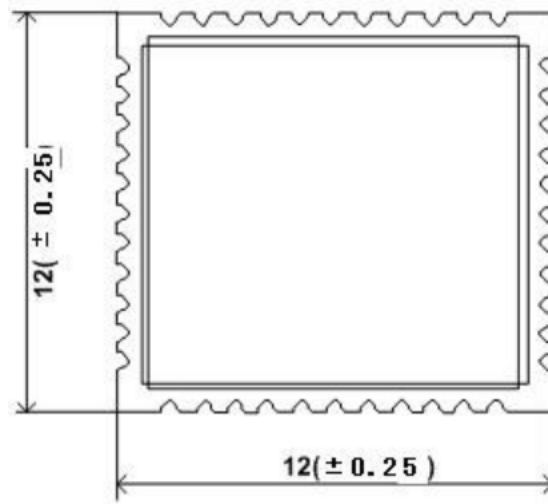


Fig 5(a) Module Dimension L × W

Product Dimension:

- H: 2.47 (± 0.2) mm



Fig 5(b) :Product Dimension

PCB thickness : 0.8 mm

Product Height : 2.47 ± 0.2 mm

Weight : 0.41(± 0.1)g

Recommended Reflow Profile :

Referred to IPC/JEDEC standard

Peak Temperature : $\leq 260^{\circ}\text{C}$

Time within 5°C of peak

temperature : $\geq 10\text{s}$

Number of Times : ≤ 2 times

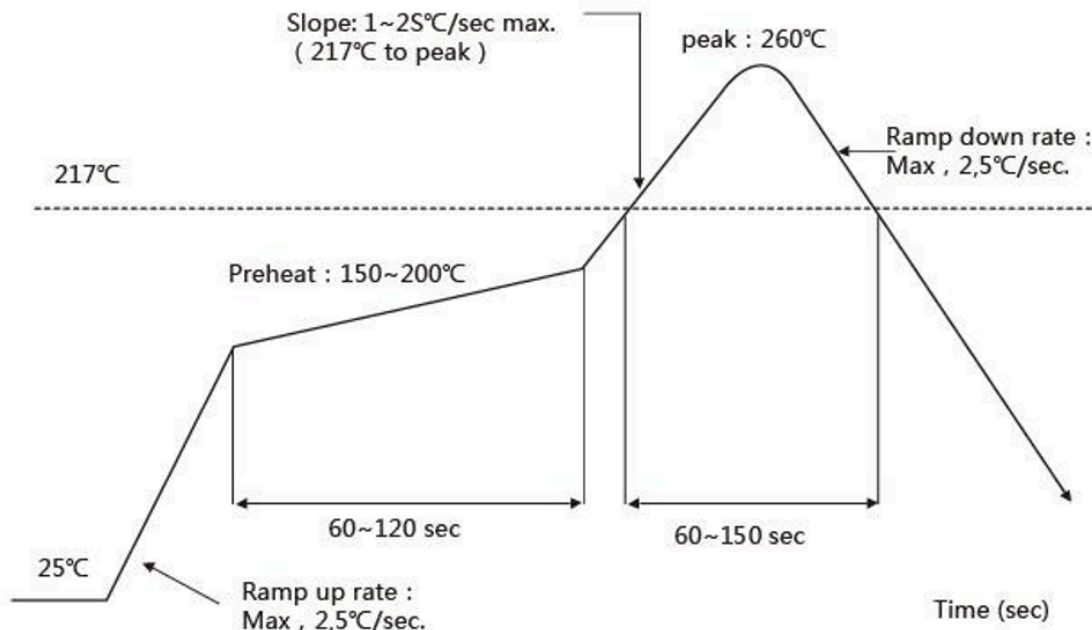
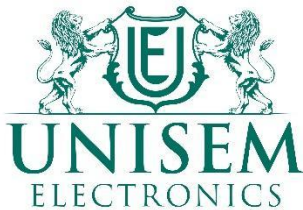


Fig 6 : Recommended Reflow Profile

For ordering USE_8800_LH please contact us on info@unisemindia.com.



Disclaimer and Copyright Notice

Unisem India Pvt Ltd or third parties may hold intellectual property rights in the products, names, logos and designs included in this document. Copying, reproduction, or modification of this document or any part thereof is only permitted with the express written permission of Unisem India Pvt Ltd. Disclosure to third parties is permitted for clearly public documents only. The information contained herein is provided "as is". No warranty of any kind, either express or implied, is made in relation to the accuracy, reliability, fitness for a particular purpose or content of this document. This document may be revised by Unisem India Pvt Ltd at any time. For most recent documents, please visit www.unisemholding.com.